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Tribal Knowledge Deconstructed

A Three-Part Epistemological Framework for Organizational and Individual Knowledge Assets

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TABLE OF CONTENTS

ABSTRACT 1

INTRODUCTION 2

THE THREE-PART FRAMEWORK 4

- Part 1: Coping Infrastructure
- Part 2: Situated Knowledge
- Part 3: The Capability Model

THE HARD BOUNDARY 12

- What cannot be transferred
- The AI replacement fallacy
- The honest framing

IMPLICATIONS FOR ORGANIZATIONAL CONTINUITY 14

- The persistent role
- The mentor voice
- The snapshot and the living dynamic
- The capability template for hiring

IMPLICATIONS FOR INDIVIDUAL KNOWLEDGE RIGHTS 16

- A new asset class
- Asset curves
- What the individual owns
- The connection to organizational failure

CONCLUSION: THE FRAMEWORK AS STARTING POINT 18

REFERENCES 19

ABOUT THE AUTHOR 20

CONTACT 21

Abstract

The term "tribal knowledge" has become analytically useless. Over two decades of knowledge management failure have produced mountains of documentation and almost no actual knowledge transfer. This paper proposes that the problem is definitional: organizations and technologists have been treating a three-part phenomenon as if it were one thing. The result is that every initiative captures the wrong part, fails to capture the right part, and leaves practitioners and institutions alike impoverished.

This paper introduces a three-part epistemological framework that separates coping infrastructure from situated knowledge and both of these from the capability model. It argues that these three parts have different origins, different ownership claims, different asset curves, and entirely different implications for organizational continuity and individual professional sovereignty. Getting the distinction right is not an academic exercise. It is a prerequisite for any knowledge strategy that actually works.

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Introduction

Ask a chief human resources officer what tribal knowledge is, and they will describe a retention problem. Ask the IT department and they will call it a documentation gap. Ask knowledge management consultants and they will speak of undocumented processes. Ask engineers on Reddit at midnight, after the third production incident in a week, and they will use words that should not appear in a white paper. All of them are looking at the same elephant. None of them are describing the same animal.

The term itself is the problem. "Tribal knowledge" started as a placeholder for everything outside the enterprise system of record: the informal, the implicit, the human. Over time, the placeholder became the category. Organizations began launching knowledge capture initiatives, appointing knowledge officers, purchasing knowledge management platforms, and building documentation workflows, all under the shared assumption that the thing they were after was one thing. It is not.

Every knowledge management initiative that has tried to capture tribal knowledge has actually collected artifacts. It has documented the workarounds. It has cataloged the shadow spreadsheets. It has transferred the sticky notes into Confluence. Then the new person arrives, reads the Confluence page, cannot function, and the organization is baffled. The knowledge transfer failed. The documentation was thorough. What went wrong?

What went wrong is that the artifacts were never the knowledge. They were the evidence that knowledge existed. The institution captured the duct tape and left the engineer behind.

> *They have been telling you to "document your tribal knowledge" for twenty years. What they actually captured was the duct tape. They never captured the engineer.*

The reason the industry has collapsed these distinctions is structural. From the perspective of enterprise software, everything that is not in the system is equally invisible. Maria's reconciliation spreadsheet and Maria's twenty years of financial judgment are both just "the messy human stuff outside the logic." The system has no architecture for distinguishing an artifact from the intelligence that produced it, so it lumps them together under a lazy label and calls it a knowledge problem.

The evidence is consistent across decades and across technologies. McKinsey has documented for years that roughly seventy percent of enterprise transformations fail to achieve their objectives. Their own AI survey sharpens the picture: eighty-eight percent of organizations now deploy AI in at least one business function, but only a third have scaled their programs and only thirty-nine percent report any measurable impact on earnings. The machines are learning everything that is in the system. They are not learning what was never there to begin with.

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The explanations offered are always the same: poor change management, insufficient engagement, misaligned incentives. What nobody names is the structural cause underneath all of them: the knowledge that actually runs the operation was never captured, because nobody could tell the difference between the workaround and the expertise that built it. Fixing this requires a framework, not another platform. The framework comes first. Here is the one that the field has been missing.

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The Three-Part Framework

If it is not one thing, what is it? The diagram below decomposes what the industry calls "tribal knowledge" into three distinct phenomena. All three emerge from the same source: the lived experience of practitioners doing real work in real systems. But they spiral outward in fundamentally different directions, with different ownership claims, different asset curves, and different consequences when someone walks out the door.

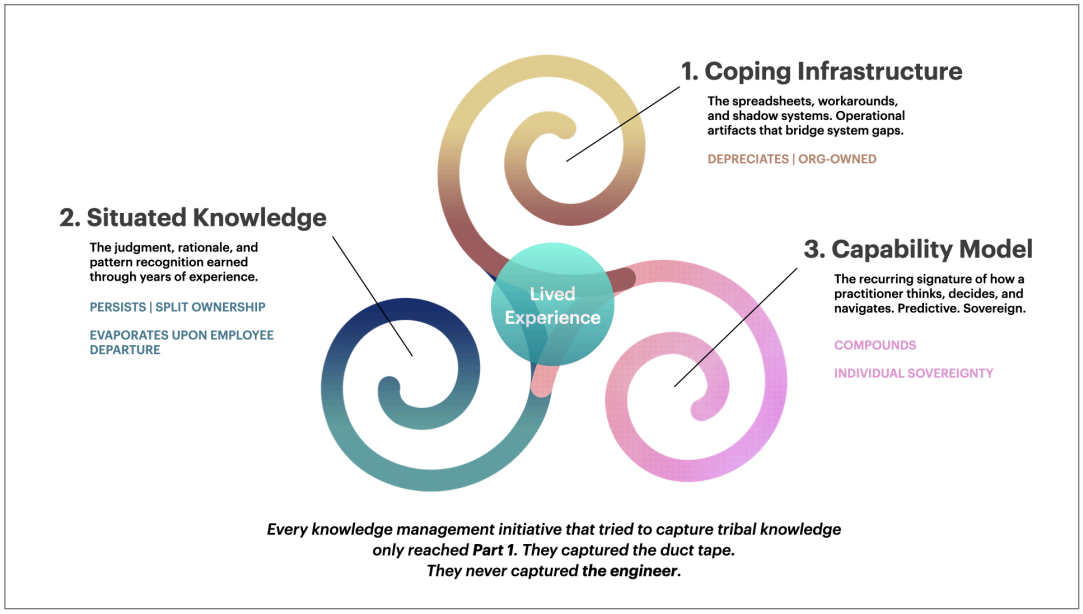

Fig. 1 – Tribal Knowledge Deconstruction

1. Coping Infrastructure is the outermost and most visible part. These are the spreadsheets, workarounds, and shadow systems that teams build to bridge the gap between what enterprise software provides and what operations actually demand. They are organizational assets. They are diagnostic. And they depreciate. When you fix the root cause, the workaround should disappear.

2. Situated Knowledge is the practiced part. This is the judgment, rationale, and pattern recognition that a practitioner earns through years of experience in a specific domain. It spans employer-specific insight (how this role works at THIS company), domain-portable expertise (clinical judgment that travels across employers), and meta-cognitive awareness (how the practitioner learns and knows what they do not know). Ownership here is split: some of it serves the organization, but the expertise itself belongs to the individual who developed it. It persists across roles, but it evaporates from the organization the day that person leaves.

3. The Capability Model is the deepest part. This is not any single artifact or any single episode of judgment. It is the recurring signature of how a practitioner thinks, decides, perceives, and

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navigates: the second-order pattern that emerges across many episodes of situated knowledge. It represents what someone is capable of, not merely what they have done. It is predictive, sovereign, and unlike everything else in this framework, it compounds continuously over the arc of a career. No employer created it. None can claim it.

One depreciates. One persists. One compounds. The sections that follow walk through each part in detail.

Part 1: Coping Infrastructure

What it is

Walk into any department that has been operating for more than two years and look at the file folders, the shared drives, the desktops of the people who have been there longest. You will find reconciliation spreadsheets. You will find shadow databases that exist because the official system cannot produce the reports the team actually needs. You will find personal scripts someone wrote to automate the step the ERP was supposed to handle but never did. You will find handwritten setup notes taped to the side of machines, annotated checklists that bear no resemblance to the official SOP, escalation paths that exist nowhere in the org chart.

This is coping infrastructure. It is the visible residue of organizational knowledge: operational artifacts created by practitioners to bridge the gap between what enterprise systems provide and what operational reality demands. It is objective, separable from its creator, and inventorable. You can put it in a spreadsheet. You can hand it to a new person. You can, in principle, delete it.

Why it matters

Coping infrastructure is diagnostic before it is anything else. Its existence tells you something important: where your enterprise systems fail to match operational reality. A department with thirty shadow spreadsheets is a department whose ERP does not work as designed. The spreadsheets are symptoms. The disease is somewhere upstream.

Consider a few specific forms from real practice. A finance team at year-end close maintains a reconciliation spreadsheet that takes two experienced accountants three hours to run. The official system should handle this. It does not, because two acquired companies were never fully integrated into the chart of accounts. The spreadsheet is a workaround for an IT integration project that has been on the backlog for four years. It is organizational bandaids over an organizational wound.

Or consider the setup notes taped to the side of a CNC machine on a manufacturing floor. The machine's official documentation says to run the initial calibration sequence, then proceed to production. What the note says is: run the calibration sequence, wait fourteen minutes even if the system says it is ready, because the spindle temperature sensor reads false positives in the first twelve minutes after a cold start and three batches of parts got scrapped before anyone figured that out. The note is not in the manual. It never will be.

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Or the IT incident responder who has a personal note file of log query strings that actually work, because the documentation was written for a previous version of the logging infrastructure and was never updated, and the correct queries took him eight months of trial and error to develop. Every new person on the team asks him for the queries. He shares them. He has been meaning to put them in the wiki for two years.

What it is not

This part is useful for modernization diagnostics. It tells you where systems need work. It can be documented, transferred, and acted upon. It is also mortal. When you fix the root cause, the workaround should disappear. Systems change, processes evolve, the adapted infrastructure becomes obsolete.

> *The spreadsheet is a fossil. The living organism was Maria's understanding of why the system could not handle those exceptions in the first place.*

Most knowledge management initiatives stop here. They capture the fossils, call it knowledge transfer, and wonder why the new person still cannot function. They have cataloged the duct tape, not the engineering judgment that produced it and that knows, in real time, when to apply it and when the situation is different enough that the duct tape will not hold.

Ownership of this part is clear: it was created in organizational context to serve organizational operations. It belongs to the institution. Which makes it the easiest part to think about and, unfortunately, the least valuable one to capture.

Part 2: Situated Knowledge

The part that actually loses

When a senior practitioner walks out the door, the organization is not grieving because they are losing access to a set of spreadsheets. They are grieving because they are losing something harder to name: the person's judgment. Their pattern recognition. Their contextual awareness built up over years of consequential decisions in this specific environment, with these specific systems, these specific teams, this specific institutional history.

That is situated knowledge. It is the judgment, rationale, and cognitive awareness that practitioners generate through sustained practice in a real environment. It is more sophisticated than coping infrastructure, but it shares one crucial property with it: once externalized, it becomes an artifact. A captured rationale is a representation of a cognitive process, not the process itself. It is the best map we can produce of the territory. But it is not the territory.

Understanding this distinction is critical. When someone asks a practitioner "why do you do it that way?" and the practitioner articulates their reasoning, what gets captured is a linguistic

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model of their cognitive process. It is genuinely valuable. It is also incomplete in ways that matter when the situation is novel and the documentation does not cover it.

Three sub-categories, three ownership structures

Situated knowledge is not uniform. It splits into three sub-categories that have fundamentally different ownership implications, and conflating them has been the source of enormous institutional injustice and practical inefficiency.

Employer-specific situated knowledge

How this role works at THIS company, with THESE systems, THESE team dynamics, THESE customers. The specific organizational context that generates it makes the ownership claim relatively clear: this knowledge belongs to the context that shaped it. It was produced in service of organizational operations and is largely inseparable from the organizational environment in which it was developed. It should persist in the role even as individuals pass through.

Domain-portable situated knowledge

Clinical judgment developed across a nursing career that spans four hospital systems. Engineering heuristics for system reliability built across fifteen years and six employers. Regulatory interpretation instincts that apply across industries because the underlying logic is the same. This part was not created by any single employer. It was created by the practitioner, through the practitioner, across a career that preceded any individual employer and will survive all of them.

No employer funded this knowledge from scratch. No employer should be able to claim it when a practitioner leaves. The practitioner brought domain expertise into the role, developed it further through practice, and will carry it to the next engagement. The domain-portable part belongs to the individual.

Meta-cognitive situated knowledge

How this practitioner learns. How they recognize the limits of their own knowledge. How they adapt when the environment changes. How they know what they do not know. This is the most purely personal category and, over a career, the most valuable. It is the practitioner's epistemological operating system: the meta-capacity that determines how quickly and how deeply they develop in any new environment.

> *The sovereignty claim is not "I own everything I produced at work." It is "I own the expertise I developed through working, and no employer should be able to strip that from me when I leave."*

The persistent role: a conceptual sketch

One implication of this part deserves attention, because it suggests something new is possible. The employer-specific situated knowledge in a role does not have to reset to zero every time someone leaves. It has been accumulating with every practitioner who has occupied that role. The institutional memory is there, distributed across the recollections and documents and

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informal practices of everyone who has passed through. What has been missing is an architecture for it to persist.

Imagine a knowledge structure associated with a role rather than any individual occupant. Each person who holds the role contributes to it. Each successor inherits the accumulated context of everyone who came before. The incoming person does not start from scratch. They inherit a head start. And when they leave, they leave the role richer than they found it.

This possibility is explored in depth later in this paper.

Part 3: The Capability Model

The part nobody is capturing

The first two parts are artifacts. One part is operational artifacts, created to bridge system gaps. The other is cognitive artifacts, created when practitioners articulate their reasoning. Both are produced by the same underlying phenomenon, and neither of them is that phenomenon.

The capability model is different in kind. It is not an artifact at all. It is a derived inference: what emerges when you look across many episodes of a practitioner's captured knowledge and ask what patterns appear consistently. Not what they did once, but what they do consistently. Not what they said about one decision, but how they structure decisions across hundreds of situations over years of practice.

A single captured episode, a situation described with its rationale, is a first-order artifact. The capability model is second-order: it is computed from the patterns across many such episodes. It is the recurring cognitive signature of how this practitioner thinks, perceives, decides, and navigates. It is what a person IS CAPABLE OF, evidenced by the patterns in everything they have produced.

This is the part that has never been captured at scale. Not because the data does not exist, but because nobody has had both the methodology to elicit it and the architecture to derive it.

The Fourteen Dimensions

The capability model organizes around fourteen recurring dimensions of practitioner cognition. Each dimension is observable across multiple captured episodes. Together they compose a cognitive portrait that is more predictive than any resume, more honest than any interview, and more durable than any reference.

1. Decision Patterns

How does this practitioner consistently weigh competing priorities across many situations? Accuracy over throughput? Stability over speed? Individual situations reveal choices. Patterns across hundreds of situations reveal the underlying value structure that shapes every choice.

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2. Diagnostic Signatures

When encountering an unfamiliar problem, what is their consistent decomposition strategy? Start from boundary conditions and work inward? Start from the most recent change and work outward? The data: recurring sequence patterns in how problems get decomposed across every domain they have worked in.

3. Contextual Trigger Recognition

What does this practitioner notice that others miss? The warehouse operations expert who knows, from ambient floor signals, that the afternoon's load will cause problems before it begins. The security monitor who reads three individually innocuous log entries as the signature of a coordinated attack. The clinical practitioner whose perception of patient deterioration precedes any measurable change. The data: a catalog of early indicators they reference, revealing the perceptual model built through thousands of consequential iterations.

4. Boundary Judgment

When does this person follow procedure and when do they deviate, and on what basis? The data: a conditional map of when they adhere versus improvise, with the rationale captured. This dimension is particularly rich because boundary judgment is where expertise becomes visible: novices follow rules or break them randomly; experts know exactly which rules exist to prevent which specific failures, and they know when the situation is different enough that the rule does not apply.

5. Transfer Reasoning

How do they apply knowledge from one domain to another? What analogies do they reach for when facing something genuinely new? This dimension maps prior experience that gets referenced when reasoning about novel problems, revealing the practitioner's capacity to generate insight at the edges of their knowledge.

6. Influence Architecture

How do they move other people? Lead with data versus relationship? Build coalitions or go direct? Do they adapt their approach to different audiences or apply the same model regardless? The data: patterns across every situation where they needed to persuade, align, or direct.

7. Collaborative Cognition

How do they think WITH others? Do they build on contributions or redirect them? Create space for productive dissent or drive to consensus? Do they elevate the thinking of the people around them, or contain it? The difference between someone whose expertise multiplies through their team and someone whose expertise stays contained within themselves.

8. Uncertainty Navigation

Their actual strategy when operating without complete information. Do they narrow scope to what is knowable? Construct provisional frameworks and test them? Make decisive moves on partial signals while remaining genuinely open to revision? This is the dimension that most

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reliably separates experienced from inexperienced practitioners, because it cannot be performed in an interview.

9. Organizational Proprioception

Their sense of where they sit in a system. Awareness of who is affected, where pressure points are, which relationships are load-bearing, which political dynamics shape what is possible. The sophistication of this awareness: one level up or three levels out. The practitioner who knows, before a meeting happens, which stakeholder will surface the objection that kills the initiative.

10. Ethical Reasoning Under Pressure

Not self-reported ethics, but the structure of actual reasoning in dilemma episodes. When the expedient path and the right path diverge, how do they navigate? What do they refuse to trade away? How do they find paths that honor their standards without destroying the relationships that make their work possible? Character made visible through the evidence of actual decisions under actual pressure.

11. Cognitive Performance Under Stress

How does their cognition change under high-stakes conditions? Do they sharpen or narrow? The contrast between routine and high-stakes reasoning is itself a data point. Some practitioners perform at their peak when the stakes are highest. Some narrow catastrophically. Most fall somewhere between. The pattern is consistently visible across episodes.

12. Developmental Trajectory

Across episodes from different periods of a career, how has their reasoning changed? Can you observe qualitative leaps? Is the rate of development accelerating, stabilizing, or plateauing? The shape of cognitive development over time is a predictor of future growth that no interview can produce and no resume can encode.

13. Mentorship and Knowledge Propagation

Do they teach principles or give answers? Create learning conditions or solve problems themselves? Whether their expertise multiplies through people around them or stays contained is a property of how they think about their relationship to knowledge, not a personality trait. The data: patterns in how they engage with less experienced colleagues across every episode where that dynamic appears.

14. Sensemaking Narrative Structure

How do they construct meaning from experience? Do they see systems or individuals? Patterns or incidents? Cause or coincidence? This is the meta-dimension, the lens through which all other capabilities are organized. Two practitioners with identical technical knowledge will produce fundamentally different capability profiles if their sensemaking structures differ, because the sensemaking structure determines what data they register, how they organize it, and what conclusions they draw from it.

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What this is not

The capability model is not a personality assessment. It is not a competency framework built from job descriptions. It is not a self-reported skills inventory. All of those instruments start from what someone says about themselves. The capability model is derived from what they demonstrate, across real episodes, under real conditions, over time.

Consider the automotive designer analogy. Nobody hires a senior designer because of the specific cars they designed at a previous employer. Those designs belong to that company. You hire them because the body of work demonstrates a capability: an integrated set of judgment, aesthetic sensibility, engineering intuition, and creative problem-solving that you believe they can apply to YOUR problems. What the capability model captures is not "here are the answers this person produced." It is "here is what this person is capable of, evidenced by the patterns across everything they have produced."

Compare it to existing representations. A resume is a self-reported list of positions held. An interview is a ninety-minute performance under artificial conditions, with decades of research showing poor predictive validity for actual job performance. References are other people's impressions filtered through relationship dynamics and recency bias. A job description is an aspirational fiction written before anyone knew what the role would actually require. The capability model is an evidence-based inference drawn from situated, rationale-rich data across many real episodes. It is the difference between reading a pilot's logbook and watching them fly in a simulator across fifty different emergency scenarios.

> *What gets licensed is not the knowledge. It is the demonstrated capability to generate knowledge. That is what has durable market value, and that is what the individual should own.*

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The Hard Boundary

What cannot be transferred

The framework above is analytically useful and commercially important. But intellectual honesty requires naming what none of it captures.

A senior practitioner does not just know things. They perceive differently. Their attention is trained. They literally notice signals that a less experienced person does not register: not because they are more intelligent, but because their perceptual system has been shaped by thousands of iterations of exposure and consequence. The manufacturing floor expert who hears a machine cycling wrong before any instrument shows a reading. The clinical practitioner whose sense of a patient's trajectory precedes any measurable deterioration. The financial investigator who reads a transaction pattern as anomalous before they can articulate why.

This perceptual capacity developed through embodied experience. It required real stakes. It required the cost of being wrong. Those shaping forces are not transferable through documentation, because documentation does not have stakes. Reading about a decision is not the same cognitive event as making it when the consequences are real.

The AI replacement fallacy

The implicit assumption behind most AI-driven knowledge work is that if you capture enough artifacts, including rationale, you can reconstruct the generative capacity computationally. This assumption confuses the map for the territory in a way that has measurable consequences.

A 2021 study of clinical decision-making made this visible in medical terms. Physicians were given chest X-rays along with AI-generated diagnostic advice, some of which was deliberately inaccurate. The results were striking. Non-expert physicians could not reliably distinguish accurate advice from inaccurate advice. But the more troubling finding was that even expert radiologists showed susceptibility to wrong AI recommendations, deferring to the system's assessment when their own clinical judgment was correct. The AI was not designed to collaborate. It offered a diagnostic recommendation with no explanatory surface, no mechanism for the clinician to interrogate its reasoning or weigh it against their own. The result was that human clinical judgment, the very expertise the system was supposed to augment, was degraded by proximity to it.

AI systems designed to automate rather than collaborate degrade the expertise they depend on. MIT economist David Autor has framed the larger pattern: the future value of AI lies in capturing use-case-specific tacit knowledge, in service of human expertise rather than in replacement of it. Black-box AI systems that automate without explanation risk destroying the process knowledge that makes expert performance possible in the first place.

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The machines never ask. So they never learn how work actually gets done. And the people who work alongside them stop practicing the judgment that made them experts.

The honest framing

What the three-part framework enables is knowledge transfer at unprecedented depth. Part one: the coping infrastructure that reveals where systems fail operational reality. Part two: the situated knowledge and rationale that experienced practitioners carry, organized into employer-specific and portable components. Part three: the derived capability model that describes what someone is actually capable of, evidenced across many episodes of real practice.

What it does not enable, and what no system will ever enable, is the transfer of the living engine itself. The embodied cognitive apparatus that generates new knowledge in genuinely novel situations requires a human to operate it. It developed through years of consequential exposure and cannot be reconstituted from its output.

This is the intellectual honesty move. It is also the strongest commercial positioning available. The companies claiming AI can replace practitioner cognition are making a promise they cannot keep, and the failures are already accumulating.

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Implications for Organizational Continuity

Once you separate the three parts, entirely new approaches to organizational continuity become possible. Not incrementally new. Fundamentally new.

The persistent role

Every time someone leaves a role, the organization loses whatever that person knew that was specific to that role in that context. The incoming person starts from scratch on the employer-specific situated knowledge, regardless of how experienced they are in the domain. The onboarding curve is universal, predictable, and enormous. Organizations have accepted this as a cost of doing business because no alternative architecture existed.

The three-part framework suggests a different possibility. If the employer-specific situated knowledge in a role can be captured, structured, and persisted, then each occupant of a role contributes to a knowledge structure that accumulates over time. The incoming person inherits context, institutional memory, and a massive head start. The role becomes richer with each occupant rather than resetting with each departure.

This is not a description of documentation. Documentation captures part one artifacts. What persists in the role is the situated knowledge and rationale of every predecessor: how they navigated the specific systems, the specific team dynamics, the specific organizational pressures of this environment. That is qualitatively different from a process manual, and it is qualitatively different from anything that has existed before.

The mentor voice

Consider what it would mean for a new practitioner to be able to engage with how their predecessor thought. Not read their notes. Engage with their reasoning, in real time, when facing a specific situation. Not "here is the documentation for this process" but "here is how Marcus navigated the tension between the finance team and the operations team on the quarterly close, and here is the reasoning behind it."

This is not an instruction manual. An instruction manual tells you what to do. This is a mentor who is no longer there: advisory, contextually relevant, initiated by the new person when they are stuck. The difference between having a knowledgeable colleague to think with and having a policy document to consult is the difference between learning and compliance.

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The snapshot and the living dynamic

When a predecessor leaves, the organizational knowledge associated with their role snapshots. It begins to age. Systems change, team dynamics shift, the snapshot becomes increasingly historical. When the new person arrives and begins contributing their own situated knowledge and rationale, the organizational knowledge comes back to life, enriched by the new perspective.

Meanwhile, the individual's capability model does the opposite. It keeps compounding. Every new role adds new episodes, new dimensions, new evidence of what this person is capable of. The organizational knowledge associated with their previous role ages. Their own capability model appreciates.

The capability template for hiring

Organizations universally know who their exceptional performers are. They virtually never can articulate what makes those people exceptional in a way that is transferable to a hiring decision. The job description describes what the role requires on paper. It does not describe what the most successful occupants of that role actually did, how they thought, what dimensions of their capability made the difference.

A rich capability model of an exceptional practitioner in a role provides something unprecedented: not a job description, but a demonstrated model of what success actually looked like. The organization can evaluate candidates not against keyword-matched resumes but against the specific capability dimensions that proved decisive. Not looking for a clone. Looking for evidence of similar cognitive patterns in the dimensions that mattered most in practice.

> *Every organization already knows who their star performers are. None of them can articulate what actually makes those people exceptional in a way that is transferable. The capability model makes the invisible architecture of excellence visible.*

One caveat is important here. There is a line between using a predecessor's capability model as organizational wisdom and using it as a surveillance template. The model is descriptive of one path to excellence. It is not prescriptive. A successor might achieve equivalent or better outcomes through a completely different capability profile. The organizational value is navigational, not normative.

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Implications for Individual Knowledge Rights

The capability model is not only an organizational asset. It is the individual's sovereign professional capital. The one thing that no employer funded from scratch. The one thing that no employer can claim to have created. The one thing that travels with the practitioner through every role, every transition, every chapter of a career.

A new asset class

Every hiring platform in existence trades in shallow representations: resumes, profiles, endorsements, ratings. These instruments share a common limitation: they are assertions about what someone is capable of, rather than evidence of it. The resume says the person held the title. The interview is a performance under artificial conditions. The reference is filtered through social dynamics and the specific experience of one individual in one context. The endorsement is a reciprocal gesture.

A capability model derived from deep conversational elicitation is an evidence-based representation of what a person is actually capable of, grounded in real episodes from real practice across the real complexity of their career. It is not a self-assertion. It is a derived inference from situated data. The difference is the same as the difference between a logbook entry and a flight record: one is what the pilot claims, the other is what they demonstrated.

Asset curves

Coping infrastructure, part one, is highest value at the moment of departure and declines continuously from there. Systems evolve, processes change, the workarounds become obsolete. It is a depreciating organizational asset that belongs to the institution.

The individual capability model is lowest value at career start and appreciates continuously. Every role adds episodes. Every challenge adds dimensions. Every failure, every adaptation, every novel situation navigated successfully enriches the model. The asset curve is the inverse: it compounds over a career rather than depreciating.

These are different assets with different owners and different trajectories. Treating them as the same thing, as every existing knowledge management framework does, produces policies that are both analytically wrong and practically unjust.

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What the individual owns

The individual does not own the specific artifacts they produced for their employer. The reconciliation spreadsheet belongs to the company. The runbook with the undocumented steps belongs to the company. The client escalation cheat sheet belongs to the company.

What the individual owns is the expertise they developed through producing those artifacts. The financial judgment that allowed them to build a reconciliation spreadsheet that actually worked. The systems understanding that made the runbook useful rather than dangerously wrong. The client relationship intelligence built through five years of difficult conversations. The portable parts of accumulated capability: domain knowledge and cognitive frameworks built through years of practice that no employer created and none can claim.

Jeff DeGraff's concept of "spectral accountability" frames the risk on the other end: the gradual transfer of genuine cognitive work to AI systems while humans remain as ceremonial approvers of decisions those systems pre-formed. Responsibility without authorship. The individual loses agency without ever consciously consenting to the trade. Knowledge sovereignty is the structural response: ensuring that the cognitive work of practitioners is recognized, attributed, and owned by the people who generated it.

> *What the individual owns is not any particular artifact. It is the demonstrated capability to generate knowledge in novel situations. That is what has durable market value, and that is what should be sovereign.*

The connection to organizational failure

Individual knowledge rights and organizational knowledge continuity are not competing concerns. They are the same problem viewed from different positions. When practitioners have no structure for capturing and preserving their own expertise, they leave with everything they know and no part of it persists. When organizations have no mechanism for accumulating the employer-specific situated knowledge in a role across successive occupants, they lose it with every departure.

The framework proposed here suggests that both problems have the same underlying solution architecture: a system that captures the right parts, attributes them correctly, and routes them to the right owners. Part one artifacts to the organization for systems improvement. Employer-specific situated knowledge to the organizational role for continuity. Domain-portable and meta-cognitive knowledge to the individual practitioner for lifetime professional sovereignty.

Getting this right is not a technical challenge. The technical challenges are substantial, but they are tractable. Getting this right is first a conceptual challenge: building a framework precise enough to separate what has been conflated for twenty years. That is what this paper is for.

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Conclusion: The Framework as Starting Point

The term "tribal knowledge" is a placeholder that has outlived its usefulness. It was never analytically rigorous. It served as a stand-in for everything the enterprise system could not see, which turned out to be the most important things in the organization: the judgment, the pattern recognition, the contextual intelligence, the cognitive capacity that makes the difference between an organization that learns and one that repeats its failures.

The three-part framework proposed here replaces the placeholder with a structure precise enough to act on. Coping infrastructure: the operational artifacts that reveal where systems fail and belong to the organization that generated them. Situated knowledge: the judgment and rationale that experienced practitioners carry, with employer-specific components belonging to the organizational role and portable components belonging to the individual. The capability model: the derived cognitive signature of what a practitioner is capable of, evidenced across many episodes, appreciating continuously, and belonging unambiguously to the individual.

Each part has different implications for how knowledge should be captured, preserved, attributed, and transferred. Each part has different implications for what we owe practitioners, what organizations are entitled to retain, and what AI systems can and cannot do. Getting the parts confused is not an academic error. It is the reason two decades of knowledge management investment has produced so little durable value.

This is the first paper in a series. Future papers will develop the capability model in depth, examine the role as a persistent knowledge entity, and explore the legal and economic implications of dual-ownership knowledge architecture in the age of AI. Each paper builds from the framework established here.

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3. Gaube, S., Suresh, H., Raue, M., et al. (2021). "Do as AI Say: Susceptibility in Deployment of Clinical Decision-Aids." npj Digital Medicine, 4(1), 31. Physicians receiving AI-generated diagnostic advice deferred to inaccurate recommendations and overrode their own correct clinical judgment. Demonstrates the degradation of human expertise when AI is designed to automate rather than collaborate. Referenced in The Hard Boundary.

4. McKinsey & Company. (2019). "Why Do Most Transformations Fail? A Conversation with Harry Robinson." McKinsey Transformation Practice. Research across corporate transformations finds that roughly seventy percent fail to achieve their objectives. Contributing factors include insufficient aspiration-setting, lack of organizational engagement, and inadequate investment in building capabilities. Referenced in Introduction.

5. McKinsey Global Institute. (2025). "The State of AI in 2025: Agents, Innovation, and Transformation." McKinsey and Company. 88% of organizations deploy AI; approximately one-third have scaled programs; 39% report measurable EBIT impact. The implementation gap that this framework is designed to address in part.

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About the Author

Chris Dollard is the Founder and CEO of ABRAXIS, Inc. His IT career spans more than forty years, with more than half of that time inside the enterprise systems this paper describes: Pacific Research Centre at IBM, and enterprise transformation work at Deloitte and Accenture. He also worked for four years at a legacy modernization company (bought by Dell Computer in 2012) where the solution set specifically identified tribal knowledge as an asset class most system integration companies and transformation projects missed. Since then he worked as COO at a two biosciences companies, one of which was taken public on the TSX.

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